

Assignment: AI Agent Formulation and Analysis

Selected Agent: Car Maintenance Prediction Agent

1. Problem in the Agent's Context

The main problem is to predict when a car will need maintenance or repairs. The agent needs to monitor the car's mileage, engine condition, oil level, temperature, battery condition, and other important information. It should identify possible problems early and recommend maintenance before a major breakdown occurs.

2. Problem Formulation

Textual Formulation

States: The agent's current vehicle condition, mileage, engine temperature, oil level, battery condition, tire condition, and maintenance history.

Initial State: The car is operating normally with its current mileage and maintenance information recorded in the system.

Actions: Monitor Vehicle, Check Condition, Detect Problem, Predict Maintenance, Send Alert, and Recommend Service.

Transition Model: Monitoring the car provides new information about its condition. As mileage increases or vehicle conditions change, the agent updates the vehicle's maintenance status and may recommend servicing.

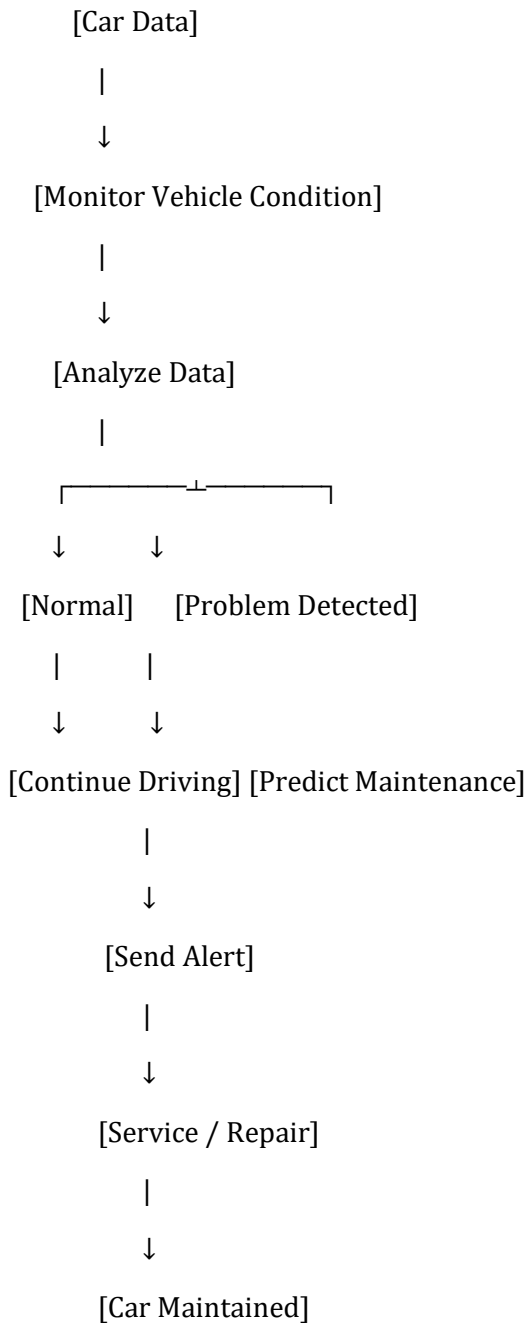
Goal Test: The car is properly maintained and potential problems are detected before they cause a major breakdown.

Path Cost: Maintenance checks and repairs require time and money. The goal is to reduce unexpected breakdowns and maintenance costs while keeping the car in good condition.

Graphical Formulation

A simple flow representing the car maintenance prediction process:

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3. PEAS Evaluation

Performance Measure: Accurate maintenance predictions, fewer unexpected breakdowns, reduced maintenance costs, vehicle safety, and timely servicing.

Environment: Roads, car engine, garage, service center, traffic, weather conditions, and vehicle components.

Actuators: Warning lights, dashboard notifications, mobile notifications, maintenance reminders, and service recommendations.

Sensors: Mileage sensor, engine temperature sensor, oil level sensor, battery sensor, tire pressure sensor, and diagnostic sensors.

4. Environment Properties

- Partially Observable
- Stochastic
- Sequential
- Dynamic
- Continuous
- Single-Agent